

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	26-06-2026
Team ID	_____
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

During the brainstorming session, the team identified challenges faced by students and educators in understanding learning difficulties. The discussion focused on building an AI-based solution capable of detecting emotions from student text and generating personalized learning support.

Step 1: Selected Problem Statement

Develop an AI-powered Learning Assistant that detects learner emotions from text using BiLSTM and BERT models and provides personalized guidance using Gemini AI.

Step 2: Brainstorm Ideas

Member / Category	Ideas
Member 1	Emotion Detection, Streamlit UI, User Login, Project Workflow
Member 2	BiLSTM, BERT, Confidence Score, Mixed Emotion Detection
Member 3	Gemini API, Personalized Responses, CSV Logging, Analytics Dashboard
Member 4	Cloud Deployment, Model Comparison, Plotly Charts, Future LMS Integration

Grouped Ideas

- Emotion Detection: NLP, BiLSTM, BERT, Mixed Emotion Detection, Confidence Score
- AI Learning Support: Gemini AI, Personalized Responses, Encouragement, Study Tips
- User Interface: Streamlit Dashboard, Login, Model Comparison
- Analytics & Storage: CSV Logging, Plotly Visualizations, Emotion History

Step 3: Idea Prioritization

High Priority	Medium Priority	Future Enhancements
Emotion Detection (BiLSTM & BERT)	CSV Logging	Voice Emotion Detection
Gemini AI Responses	Cloud Deployment	LMS Integration
Streamlit UI	Analytics Improvements	Multilingual Support

Final Selected Idea

Emotion Detection & Learning Support Engine

The team selected this idea because it combines Natural Language Processing, Deep Learning, and Generative AI to understand students' emotions and provide timely, personalized academic support.